O-Drift: Observability-Aware Off-Policy Evaluation in Evolving Recommender Systems

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Abstract. Off-policy evaluation for the current period must often rely on logs collected over many earlier periods, with only a small sample from the period itself. Common remedies (recency windows, decay, forecasting) treat every change alike and discard data that is still informative. We argue that the useful question is not how recent a period is but which mechanism changed: a change in the context distribution or in the logging policy can be corrected by reweighting, while only a change in the conditional reward biases a reweighted estimate. We propose O-Drift, which segments the history with a per-action residual test that is invariant to context and logger drift, fits joint reward models across segments, reuses older segments with a drift correction, and pools the segment estimates at minimum variance with a random-effects interval. In a pre-registered evaluation with a one-shot test split, O-Drift outperforms a cross-domain estimator that groups periods by mean reward in all 15 reward-drift settings, and replacing its residual test by mean-reward grouping causes spurious splits under context and logger drift; a second pre-registered test on fresh environments, adding the time-feature estimator OPFV, passes every criterion again. On semi-synthetic Open Bandit data, however, drift-corrected reuse is strongly biased; we report this failure and show why validation on the small current-period sample cannot repair it. On real clicks with natural drift (Open Bandit's A/B test, with Thompson sampling's own click rate as ground truth), O-Drift stays within 5% of the best of 19 baselines.

Keywords: Off-policy evaluation · Non-stationarity · Concept drift · Recommender systems

1 Introduction

Recommender systems decide whether to deploy a policy by estimating its value from logs of earlier policies, which off-policy evaluation (OPE) allows when the logs record propensities [10,8]. The logs span many weeks while users change, loggers are replaced and click behaviour moves; what is needed is the value *now*, in a target period with only a small sample.

Pooling all history gives a low-variance estimate that is biased whenever the environment has changed; using only the small target sample (the *anchor*) is unbiased but noisy. Existing remedies interpolate between the two along the time axis: sliding windows and moving averages [11], exponential decay, forecasting

the policy value as a time series [20,3], and regression-assisted estimators that reuse a model fitted on recent data [13]. All of them assume that older data is less useful because it is older.

This assumption ignores what changed. A logged record is generated by three mechanisms: the context distribution $P_t(x)$, the logging policy $\mu_t(a \mid x)$ and the conditional reward $q_t(x, a) = \mathbb{E}[R \mid x, a, t]$. The first two are nuisances that standard tools already correct: importance weights handle the logger [1], and density-ratio transport handles the contexts [21]. Only a change in q_t biases a correctly reweighted estimate. An old period whose users looked different but clicked in the same way is as informative as yesterday; a recent period after a change in click behaviour is not. Recency is a proxy for the mechanism that matters, and a poor one when contexts or loggers drift more often than rewards.

We turn this observation into an estimator, O-DRIFT (off-policy evaluation under drift). It tests for drift in the conditional reward directly, with a per-action residual statistic computed from one pooled reward model, and segments the history by binary segmentation. Under correct specification the test keeps its nominal level under any context or logger drift (Proposition 1), whereas statistics built from raw rewards, such as the mean-reward similarity used by the cross-domain estimator COPE [14], do not. Within each segment O-DRIFT pools windows with balanced multiple-logger weights; across segments it fits joint reward models with a shared base and per-segment offsets, reuses older segments after correcting their reward offset, and combines the segment estimates with the anchor at minimum variance.

A second question is what can be seen at all. Drift *between* historical windows is observed through tens of thousands of records, so even small changes in q_t are detectable; drift between the history and the target period is observed only through the anchor, so only large changes are. O-DRIFT is *observability-aware* in this sense: it segments the history where reward drift is observable, uses the anchor only for the coarse decision of which segment is closest to the target, and declines to certify an estimate when no anchor is available. The same view explains its main failure: the bias of drift-corrected reuse is too small for the anchor to observe and too large to ignore.

Our contributions are:

- a mechanism decomposition of drift for OPE, an observability analysis of which reward drift the data can reveal, and a reward-drift monitor whose size does not depend on context or logger drift (Sections 3 and 4);
- the O-DRIFT estimator, with segment-wise joint reward models, drift-corrected reuse, minimum-variance pooling and a random-effects interval (Section 4);
- a pre-registered evaluation on simulated regimes, semi-synthetic Open Bandit data [16] and Open Bandit with real clicks and natural drift, each test split run once, against 19 baselines including faithful implementations of COPE and OPFV [17], with a grouping ablation that isolates the monitor (Sections 5 and 6).

We also report a negative result in full: with a misspecified reward model and large drift, drift-corrected reuse is heavily biased, and validating reuse on the anchor or a held-out window cannot detect that bias (Section 6).

2 Related Work

Off-policy evaluation. Inverse propensity scoring [10] and the doubly robust (DR) estimator [8] are the standard tools for contextual bandit OPE; we cross-fit the reward models as in double machine learning [5]. Weighting by the mixture of several logging policies (balanced or multiple importance weighting) reduces variance [1,12], and density-ratio transport handles covariate shift [21]. O-Drift uses all three; it adds the decision of which historical data the reweighted estimate may use.

Non-stationary OPE. Jagerman et al. [11] study sliding-window and exponentially decayed estimators for non-stationary bandits. Thomas et al. [20] and Chandak et al. [3,4] forecast the future value of a policy from its past per-period estimates, and Shimizu et al. [17] estimate future values by exploiting time features shared between past and future. Liu et al. [13] reuse a reward model fitted on older data inside a DR estimator on the new data. These methods weight data by time or by a model of the trend in the value; none distinguishes the mechanism behind a change.

Cross-domain OPE. COPE [14] treats other domains as sources for a target domain, clusters domains by the similarity of their mean rewards and combines multiple importance weighting inside the target cluster with a decomposed reward model. With time windows as domains it is the closest existing estimator to ours, but its clustering statistic, the mean reward, mixes all three mechanisms; we include it as a tuned baseline and as a grouping ablation.

Estimator selection and drift detection. MAGIC [19], SLOPE [18] and adaptive weighting [22] choose among or blend estimators of increasing bias and decreasing variance (baselines here, over nested recency windows); drift detection on streams is surveyed in [9]. Our monitor is a heterogeneity test [6] on per-action residuals and our interval the DerSimonian–Laird interval [7]; the novelty is the statistic tested, not the tests. Selecting segments by a preliminary test is a pretest estimator [2], with its known risk near the threshold.

3 Problem Setting

Data. Time is divided into windows $k = 1, \dots, K$ followed by a target period T. Window k contributes logged records $\mathcal{D}_k = \{(x_i, a_i, r_i, \mu_k(a_i \mid x_i))\}_{i=1}^{n_k}$ with contexts $x \sim P_k$, actions $a \sim \mu_k(\cdot \mid x)$ from the logger deployed in window k, and binary rewards with conditional mean $q_k(x, a)$. From the target period we

have a small anchor $\mathcal{D}_T$ of $n_T \ll \sum_k n_k$ logged records and a sample $\{x_j^*\}_{j=1}^M$ of unlabelled target contexts. The estimand for a target policy π is

$$V_T(\pi) = \mathbb{E}_{x\sim P_T}\Big[\sum_a \pi(a \mid x)\, q_T(x,a)\Big]. \tag{1}$$

Three drift mechanisms. A window can differ from the target in P_k, in μ_k or in q_k. With logged propensities and a density ratio $w_k(x) = P_T(x)/P_k(x)$, the reweighted DR score of a record in window k,

$$\Gamma_k(x,a,r) = \mathbb{E}_{x'\sim P_T}\Big[\textstyle\sum_{a'} \pi(a' \mid x')\,\hat{q}(x',a')\Big] + w_k(x)\,\frac{\pi(a|x)}{\mu_k(a|x)}\big(r - \hat{q}(x,a)\big), \tag{2}$$

has expectation $\mathbb{E}_{P_T}[\sum_a \pi(a \mid x) q_k(x,a)]$ for any $\hat{q}$. It equals $V_T(\pi)$ exactly when $q_k = q_T$ on the support of π under P_T. Context and logger drift therefore cost variance but not bias; reward drift costs bias that no reweighting removes. The transfer bias of window k is $b_k = \mathbb{E}_{P_T}[\sum_a \pi(a \mid x)(q_k - q_T)(x,a)]$.

Assumptions. (A1) Logged propensities are the ones used at event time. (A2) Overlap: $\pi(a \mid x) > 0$ implies $\mu_k(a \mid x) > 0$ for the windows used, and the context ratio w_k is bounded. (A3) Reward drift is piecewise: the windows split into contiguous segments with a common q within each segment. (A4) For drift-corrected reuse, the difference between segment rewards lies in the reward-model class (on the logit scale). Our regimes violate each assumption in turn (S6 breaks A1, S7 A2, S3b and S5 A3, S8 and the misspecified cells A4) so that we can see where the method fails.

Observability of reward drift. Without an anchor, b_k is not identified from history: the logs constrain only $q_1, \ldots, q_K$, any q_T is consistent with them, and no method can certify an estimate. With an anchor, b_k is identified, but only up to the noise of the anchor estimate, whose standard deviation is of order $\sigma/\sqrt{n_T}$. Biases below this level cannot be detected from the anchor, however the test is built: a level-α test with power $1-\beta$ needs $|b_k| \gtrsim (z_{\alpha/2} + z_\beta)\,\sigma/\sqrt{n_T}$. We call drift *observable* when it exceeds this threshold for the sample that must reveal it. Drift between two historical windows is judged on n_k records and is observable down to small sizes; drift between the history and the target is judged on the n_T anchor records and is observable only when large; this asymmetry drives the design and its failures.

4 Method

O-Drift follows the observability asymmetry of Section 3: decisions that need to see small reward changes are made on history, and the anchor is used only where a coarse decision suffices. It has four steps: segment the history by reward drift, fit joint reward models, form one drift-corrected estimate per segment, and pool them with the anchor.

4.1 A reward-drift monitor

We fit one pooled per-action logistic model $\tilde{q}$ with linear and quadratic context terms on all historical windows. For a block B of windows and an action a, let $m_B(a)$ be the mean residual $r_i - \tilde{q}(x_i, a)$ over records in B with $a_i = a$, and $v_B(a)$ its binomial variance estimate. Two adjacent blocks L, R are compared by

$$\chi^2_{LR} = \sum_a \frac{\big(m_L(a) - m_R(a)\big)^2}{v_L(a) + v_R(a)}, \tag{3}$$

referred to a χ^2 distribution with one degree of freedom per action with enough records. Binary segmentation splits the history at the most significant boundary while the p-value is below α, giving contiguous segments $S_1, \ldots, S_J$ ordered from newest to oldest.

Proposition 1 (Invariance to context and logger drift). *Suppose $q_k = q$ for all windows in $L \cup R$ and $\tilde{q} \to q$. Then for every action a, $\mathbb{E}[m_L(a) - m_R(a)] \to 0$ whatever the context laws P_k and loggers μ_k, so (3) is asymptotically χ^2 and the test has level α under any context or logger drift. The mean-reward statistic $\bar{r}_L - \bar{r}_R$ and the per-action raw-reward statistic do not have this property.*

Proof (sketch). In window k, $\mathbb{E}[R - q(X, a) \mid A = a, k] = \mathbb{E}_{P_k(x|a)}[q_k(x, a) - q(x, a)] = 0$, because $P_k(x \mid a) \propto P_k(x)\mu_k(a \mid x)$ only reweights a function that is identically zero. Only $q_k \neq q$ moves $m_B(a)$. In contrast, $\mathbb{E}[R \mid k] = \sum_a \int P_k(x)\mu_k(a \mid x)q(x, a)\, dx$ changes with P_k and μ_k when q is fixed, which gives false alarms; and a reward change can be cancelled by an opposite change of context or logger, which gives missed drift. The per-action raw mean $\mathbb{E}_{P_k(x|a)}[q(x, a)]$ removes the action mix but still depends on P_k and on μ_k as a function of x.

If q lies outside the model class, the residual has conditional mean $\mathbb{E}_{P_k(x|a)}[q - \tilde{q}]$, which depends on P_k; context drift can then trigger the monitor. The richer the model class, the smaller this effect.

4.2 Joint segment reward models

Reward models for estimation are cross-fitted [5] per-action logistic regressions on a context basis $\phi(x)$. Across segments we fit one joint model

$$\operatorname{logit} \hat{q}_j(x, a) = \phi(x)^\top \beta_a + \phi(x)^\top \delta_{j,a}, \qquad \delta_{1,a} = 0, \tag{4}$$

a shared base plus per-segment offsets, with an L_2 penalty on the offsets scaled by $1/s^2$. The scale $s \leq 1$ acts as a ridge prior on drift size and interpolates between fully corrected ($s = 1$) and uncorrected reuse ($s \to 0$). The basis is chosen per episode between linear features and $D = 32$ random Fourier features [15] by held-out log-loss of the pooled model; the same basis is used by every method.

4.3 Reference segment and drift-corrected reuse

The reference segment is the newest segment whose uncorrected estimate the anchor does not contradict at $|z| \le z_{\mathrm{anc}}$; if all are contradicted the anchor is used alone. Each segment S_j contributes

$$\hat{V}_j = \frac{1}{M}\sum_{m=1}^{M}\sum_{a}\pi(a \mid x_m^*)\hat{q}_{\mathrm{ref}}(x_m^*, a) + \frac{1}{n_j}\sum_{i\in S_j} W_i\left(r_i - \hat{q}_j(x_i, a_i)\right), \tag{5}$$

$$W_i = \frac{\pi(a_i \mid x_i)}{\sum_{k\in S_j}\frac{n_k}{n_j}\mu_k(a_i \mid x_i)/w_k(x_i)}. \tag{6}$$

The direct-method term uses the reference model, so it targets q_{ref}; the correction term uses the segment's own model, so it removes the segment's own model error. W_i is the balanced importance weight from the target to the segment's mixture of loggers and context laws [1,12], with w_k estimated by a probabilistic classifier. For the reference segment $\hat{q}_j = \hat{q}_{\mathrm{ref}}$ and (5) is the usual balanced DR estimate. For an older segment, the difference $\hat{q}_{\mathrm{ref}} - \hat{q}_j$ is exactly the fitted reward drift, so (5) transfers the segment's evidence to the reference reward when (A4) holds.

4.4 Pooling and interval

The anchor DR estimate joins the segment estimates, and the weights $\hat{\alpha} = \arg\min_{\alpha\ge 0,\,\mathbf{1}^\top\alpha=1}\alpha^\top\hat{\Sigma}\alpha$ minimise the estimated variance, where $\hat{\Sigma}$ accounts for the target contexts shared by all direct-method terms. Because bias may remain, the reported interval is DerSimonian–Laird [7], inflating the standard error by the between-component heterogeneity $\hat{\tau}^2$; a sample-split interval with an explicit bias allowance is reported as a certified but wide alternative. Without an anchor O-Drift returns an estimate flagged as not certified.

Settings. $\alpha = 10^{-4}$, $z_{\mathrm{anc}} = 5$ and $s = 0.25$ were tuned on development data (Section 5).

5 Experimental Setup

Simulated regimes. Each environment has $d = 5$ Gaussian contexts, 10 actions and per-action logistic rewards with a linear part and a nonlinear part of weight 0.5 ("misspecification 0.5") or 3.0. The level 3.0 was fixed by matching the nonlinear share of logit variance in Open Bandit (0.43 vs. 0.41) before any result at that level was seen. The history has $K = 12$ windows of 5,000 records; the anchor has 1,000 records and there are 10,000 target contexts. Ground truth is exact over actions and Monte Carlo over 10^6 contexts. The regimes isolate one mechanism at a time: S0 stationary; S1 context drift growing with age; S2 logger switches A→B→C; S3a abrupt reward drift in the older half; S3b gradual reward drift towards the target; S4 mixed (oldest third reward-drifted, middle

third context-shifted, newest third poorly explored); S5 seasonal context and reward drift; S6 wrong logged propensities (violates A1); S7 recent windows that never play three actions (violates A2); S8 reward drift along an out-of-class interaction, S8b the same without an in-class component (violate A4). Intensities are given with each regime in Table 1. Four candidate policies of increasing noise around the optimal policy are evaluated in every episode.

Open Bandit semi-synthetic data. We use the uniformly random logging data of the Men campaign of the Open Bandit Dataset [16] (452,949 records, 34 items). Real user features (8 principal components), in real time order, form 13 equal blocks (12 history windows and the target). The base reward is a per-action logistic model with linear and quadratic terms fitted to the real clicks and shifted to a mean click rate of 0.05; actions, propensities, rewards and the injected logger and reward drift are synthetic. Because the generating model is quadratic, the methods' reward models are misspecified. These data test robustness, not natural drift.

Open Bandit with real rewards. To test natural drift we use every campaign of the dataset (Men, Women, All; 34, 46 and 80 items) with nothing synthetic. The uniformly random log (0.45–1.37 million impressions over seven days) is cut into 19 time blocks of about 8.8 hours. A task takes block $b \in \{12, \ldots, 18\}$ as the target and the 12 blocks before it as history (5,000 records per window, an anchor of 1,000 and 10,000 target contexts drawn from block b). The target policy is the Bernoulli Thompson Sampling (BTS) policy that ran in the same A/B test, represented by its item frequencies per position in block b; the ground truth is its on-policy click rate in that block (0.2–0.7 million impressions, standard error $\leq 0.2 \times 10^{-3}$). Contexts are 8 principal components of the user features and the position. Interactions between items shown together are ignored, as in [16]. 21 tasks $\times$ 20 episodes.

Baselines. All share the reward models and basis rule: pooled, anchor-only, recent-k and decayed DR [11]; minimum-variance pooling; transport DR and its recent, decayed and balanced-decayed variants; regression-assisted DR [13]; SLOPE [18] and MAGIC [19] over nested windows; forecasting [20]; time-aware DR (polynomial time trend in the reward model); balanced DR [1]; an anchor-validated window selector; COPE [14] (windows as domains, anchor as target, cluster size tuned); OPFV [17], which importance-weights the records sharing the target's time feature and chooses that feature per episode from recency and periodic candidates (including the true seasonal period) by its bias–variance rule; and monitor-select, DR on the newest monitor segment plus the anchor without reuse. COPE and OPFV follow the estimators of the original papers (their Eqs. 9–10 and 4, 7) with our shared reward-model class.

Protocol. The protocol (`configs/protocol.yaml` in the code) was fixed before tuning. Every method family, baselines included, is tuned on a development split (four environments, 200 episodes per cell) by minimising the mean over

Table 1. Simulator, one-shot tests (300 episodes × 4 policies per cell; RMSE $\times 10^{-3}$). O-DRIFT, v1 (no offset shrinkage) and the best tuned baseline of the cell; ratio to it and coverage of the 95% interval. Mechanism checks: ratio to tuned COPE, ratio of the blind (mean-reward) grouping to O-DRIFT, share of episodes split by the residual monitor (R) and the blind statistic (B). Replication (v6, fresh environments): ratio to the best tuned baseline and to OPFV. *: paired 95% interval of the RMSE difference excludes 0.

			Accuracy						Mechanism checks				Replication	
Reg.	Int.	Mis.	O-DRIFT	v1	Best baseline	RMSE	Ratio	Cov.	COPE	Blind	R	B	Ratio	OPFV
S0	0	0.5	5.4	5.4	Min-var DR	5.1	1.05*	0.93	1.00	1.00	0.00	0.00	1.01	0.18*
S1	0.5	0.5	5.5	5.5	COPE	5.5	1.00	0.95	1.00	1.07*	0.00	0.22	1.00	0.19*
S1	2	0.5	8.0	8.0	COPE	7.8	1.02	0.93	1.02	1.07*	0.00	0.61	1.03*	0.27*
S2	0	0.5	5.1	5.1	Balanced DR	4.7	1.07*	0.95	1.01	1.13*	0.00	0.52	1.02	0.17*
S3a	0.25	0.5	6.4	6.5	Monitor-select DR	7.0	0.93*	0.89	0.52*	1.56*	1.00	0.29	0.84*	0.21*
S3a	1	0.5	9.8	6.5	Monitor-select DR	7.0	1.40*	0.83	0.25*	1.03	1.00	0.99	1.21*	0.27*
S3b	0.25	0.5	10.9	10.8	Time-DR	8.5	1.28*	0.63	0.82*	1.21*	0.64	0.02	1.00	0.28*
S3b	1	0.5	26.6	18.4	Time-DR	8.8	3.02*	0.27	0.60*	1.14*	1.00	0.82	2.49*	0.71*
S4	0.25	0.5	7.5	8.0	Balanced DR	8.6	0.86*	0.95	0.62*	1.25*	1.00	0.88	0.88*	0.23*
S4	1	0.5	14.1	14.8	T-recent DR	21.7	0.65*	0.78	0.32*	1.02	1.00	1.00	1.01	0.55*
S5	0.5	0.5	21.5	20.9	Decay DR	20.4	1.05*	0.34	0.93*	1.06*	0.99	0.72	0.93*	0.49*
S5	1	0.5	34.7	28.9	SLOPE	36.0	0.96	0.22	0.84*	1.13*	1.00	0.90	0.97*	0.77*
S6	0	0.5	5.9	5.9	Balanced DR	5.3	1.13*	0.94	1.07*	1.08*	0.00	0.52	0.99	0.19*
S7	0	0.5	7.4	7.4	Min-var DR	4.9	1.52*	0.94	1.30*	1.02*	0.00	1.00	1.41*	0.23*
S8	0.5	0.5	6.7	6.7	Monitor-select DR	7.0	0.96	0.88	0.42*	1.58*	1.00	0.51	0.93*	0.22*
S8	1	0.5	6.9	6.8	Monitor-select DR	7.0	0.98	0.86	0.24*	2.24*	1.00	0.50	0.96	0.22*
S8b	1	0.5	6.7	6.7	Monitor-select DR	7.0	0.96	0.89	0.49*	1.23*	1.00	0.65	0.95	0.23*
S0	0	3	4.9	4.9	Pooled DR	4.5	1.08*	0.95	1.00	1.00	0.00	0.00	1.01	0.16*
S2	0	3	5.0	5.0	Balanced DR	4.6	1.08*	0.95	1.00	1.26*	0.00	0.59	1.04	0.15*
S3a	1	3	6.5	6.7	Monitor-select DR	6.7	0.98	0.86	0.19*	2.35*	1.00	0.72	0.89*	0.18*
S4	1	3	17.9	25.7	TB-decay DR	22.0	0.81*	0.70	0.48*	1.03*	1.00	0.99	1.16*	0.59*
S5	1	3	23.3	26.2	Monitor-select DR	29.2	0.80*	0.33	0.66*	1.28*	1.00	0.72	0.87*	0.51*
S8	1	3	6.6	6.7	Monitor-select DR	6.7	0.99	0.86	0.22*	1.01	1.00	1.00	0.94*	0.19*

cells of log RMSE. Test splits use unseen environments (300 episodes per cell; 200 for Open Bandit) and are run once per frozen configuration. The method reported here is the third registered version; the first failed on Open Bandit, which led to the misspecification-3 cells, robustness criteria, per-episode basis, offset shrinkage, COPE and the grouping ablation. A later registration (v6) kept the method frozen and added OPFV, a replication on fresh test environments and the real-reward test, with criteria fixed before running. The registered criteria cover accuracy, coverage, support and abstention (C1–C5), robustness in the misspecification-3 and Open Bandit cells (R1: no ratio above 1.5; R2: ratio ≤ 1 in half of them), COPE (N1), the grouping ablation (N2), OPFV (N3, v6) and the real-reward test (E1: ratio ≤ 1.10); their wording is in the protocol file.

Metrics. RMSE ($\times 10^{-3}$); its ratio to the best tuned baseline with a paired bootstrap 95% interval of the difference (10,000 resamples); interval coverage; and the share of episodes split.

6 Results

Simulated regimes. Table 1 compares O-DRIFT with the best tuned baseline of each cell. Without reward drift (S0–S2) it is within 7% of it, the price of segment

models that turn out to be unnecessary. Under abrupt and mixed reward drift it wins (S3a: 7%; S4: 14% and 35%, against a different best baseline at each intensity), and the misspecification-3 cells of S3a, S4, S5 and S8 are won or tied (0.80–0.99). It loses in four places: gradual drift (S3b) violates the piecewise assumption and favours time-aware DR (1.28, 3.02); at the higher S3a intensity and in S5 the offset shrinkage tuned on development data under-corrects large drift (S3a: 1.40; v1 without shrinkage would have won, 6.5 vs. 7.0); and in S7 it rightly ignores support-deficient windows but loses to min-variance pooling (1.52). The heterogeneity-inflated interval undercovers when bias remains (S5, S3b: 0.22–0.63); the certified sample-split interval always covers but is 18 times wider.

Does the mechanism matter? Against tuned COPE, which groups windows by mean reward, O-Drift has lower RMSE in all 15 reward-drift cells (0.19–0.93, intervals excluding zero) and ties it in S0–S2; COPE is better only with faulty propensities or lost support (S6, S7: 1.07, 1.30). The grouping ablation replaces only the monitor statistic by the mean reward. The residual monitor never splits a history with stable reward (S0–S2, S6–S7), as Proposition 1 predicts; the mean-reward statistic splits 22–61% of S1 histories (context drift), 52% of S2 histories (logger switches) and every S7 history, costing 7–13% RMSE in S1 and S2. Under reward drift the blind grouping is never better and up to 2.35× worse, mainly because it misses abrupt changes (S3a, lower intensity: 29% of episodes detected, against 100%).

Replication and OPFV. The v6 registration reran the frozen method on four fresh environments (right columns of Table 1); all 14 registered criteria pass again, including R1–R2 on the misspecification-3 cells and the new check against OPFV (N3). The pattern replicates: within 4% without reward drift; wins in S3a and S4 at the lower intensity (0.84, 0.88), S5 (0.93, 0.97), S8 and the misspecification-3 cells of S3a, S5 and S8; losses under gradual drift (S3b, intensity 1: 2.49), in S3a at intensity 1 (1.21) and in S7 (1.41); COPE is beaten in all 15 reward-drift cells. Two wins did not replicate: S4 at intensity 1 (0.65 in the first test) is a tie with transport balanced decay (1.01), and S4 at misspecification 3 is a loss (1.16). OPFV has 1.3–6.7 times the RMSE of O-Drift in every cell. Its selection rule measures each feature's bias against the finest-feature estimate, which rests on the 1,000-record anchor; that noise swamps the bias term, so the choice is nearly random (the all-windows feature is chosen in 19% of S0 episodes). With the coarsest feature fixed, OPFV is time-aware DR, which is among the baselines.

Open Bandit: where drift-corrected reuse fails. Table 2 shows the pre-registered robustness test on Open Bandit. Without reward drift O-Drift matches the best baseline, but under reward drift it is 3.1–7.7 times worse, and almost all of its error is bias ($+14$ to $+68 \times 10^{-3}$). The monitor is not the cause: it finds the change in every S3a episode, and monitor-select (same segmentation, no reuse) is best in S3a and within 1.4× of the best in S4. The base reward is quadratic,

Table 2. One-shot test on semi-synthetic Open Bandit data (200 episodes per cell). RMSE and mean error (bias) $\times 10^{-3}$; v1 is O-DRIFT without offset shrinkage; monitor-select uses the same segmentation without reuse. Ratio as in Table 1.

Regime	Int.	O-DRIFT	Bias	v1	Monitor-select	Best tuned baseline	RMSE	Ratio
S0	0	3.2	+0.4	3.2	3.3	COPE	3.1	1.03
S2	0	2.8	+0.3	2.8	3.6	COPE	2.8	1.00
S3a	0.5	15.7	+13.6	7.5	4.7	Monitor-select DR	4.7	3.33*
S3a	1	36.4	+32.2	10.0	4.7	Monitor-select DR	4.7	7.74*
S4	0.5	32.0	+28.1	21.1	13.6	Time-DR	10.2	3.14*
S4	1	73.8	+66.4	31.8	13.5	T-recent DR	10.7	6.87*
S5	1	75.4	+68.0	58.5	27.3	Reg.-DR (Liu)	20.7	3.63*

Table 3. Real rewards and natural drift (Open Bandit, pre-registered test v6): RMSE ($\times 10^{-3}$) against BTS's on-policy click rate (mean 6.2, noise sd 0.14), 21 tasks $\times$ 20 episodes (416 with an estimate from every method). Bold: lowest in the row. Ratio of O-DRIFT to the best baseline: 1.05 [1.01, 1.09].

	O-DRIFT	Pooled	COPE	SLOPE	Mon.-sel.	Decay	MAGIC	Transp.	Forec.	OPFV	Anchor
All tasks	1.14	1.09	**1.09**	1.09	1.17	1.20	2.15	2.75	3.01	4.56	9.07
Men	1.49	1.37	1.38	**1.35**	1.55	1.44	2.54	3.66	4.12	4.19	7.79
Women	1.00	0.97	**0.93**	0.99	0.97	1.15	2.19	2.18	2.33	4.81	8.73
All items	**0.83**	0.86	**0.86**	0.86	0.86	0.95	1.60	2.12	2.18	4.67	10.50

the reward models linear (the basis rule chose linear features in every episode), and the drift three times the standard deviation of the base logit, so the offsets δ_j absorb model error as well as drift and the corrected segments carry a bias the anchor cannot observe. Shrinkage makes this worse (S3a, intensity 1: 36.4 vs. 10.0 without shrinkage). Robustness criteria R1–R2 therefore fail; the other ten pre-registered criteria (C1–C5, N1–N2) pass.

Real rewards and natural drift. On the real-reward Open Bandit test (Table 3) the click rate of the target policy moves between blocks (from 4.1 to 8.0×10^{-3}), but these changes are smaller than the error of any estimator at these sample sizes. As registered in advance, the monitor almost never splits (0.7% of episodes) and pooling everything is close to optimal: tuned COPE, whose cluster spans all windows, is best overall. O-DRIFT is within the registered margin (ratio 1.05, task-bootstrap interval 1.01–1.09; criterion E1 passed), is the most accurate method on the 80-item campaign (0.97, 0.94–0.99) and is 10% behind on Men. The offset shrinkage of v3 matters here: without it, O-DRIFT has RMSE 3.56×10^{-3} on Men. OPFV, whose rule picks fine time features, has four times the error of pooling. The interval of O-DRIFT covers in 75% of episodes. In 4 of 420 episodes the anchor contained no click, its plug-in variance is zero, and the pooling of O-DRIFT and the transport baselines is undefined; these episodes are excluded above. Post hoc, flooring that variance at one pseudo-click gives estimates in all 420 episodes with the same ratio (1.05).

Can reuse be validated on the data at hand? (post-hoc). On development data (52 matched episodes per cell), a gate (v4) that drops a segment differing from

the reference at the 5% level removes the bias but loses in S4 (1.32×; 1.61× at misspecification 3), and a rolling-origin back-test (v5) that uses the newest window as a pseudo-target rejects reuse in 31–54% of episodes without reward drift and loses in six of nine cells (up to 2.1×). Neither was tested. Both fail as Section 3 predicts: anchor DR has RMSE 42×10^{-3}, the decisive reuse bias is about 10×10^{-3}.

Anchor size and cost (post-hoc). With anchors of 100, 1,000 and 10,000 records, anchor-only DR follows $1/\sqrt{n_T}$ (RMSE 153, 42, 12×10^{-3}) while O-DRIFT hardly moves (S4: 14.2, 14.1, 11.1; no anchor 68.5), but under gradual or seasonal drift (S3b, S5) the 10,000-record anchor alone wins (11.9 vs. 23.5, 29.3): reuse bias does not shrink with n_T. On one core, O-DRIFT takes 1.0 s per episode and policy in the simulator and 1.5 s on Open Bandit, against 0.6 and 0.3 s for COPE and 0.5 and 0.1 s for pooled DR; the monitor and the joint segment fits are the extra cost.

7 Limitations and Conclusion

For OPE under drift, which mechanism changed a window matters more than its age, and a drift decision is only as good as the data that can observe it. A residual test on the conditional reward does not mistake context or logger drift for reward drift, which in our pre-registered evaluation separated O-DRIFT from mean-reward grouping in every reward-drift setting.

The limitations are as important as the result. First, drift-corrected reuse depends on the reward-drift difference lying in the model class (A4); on semi-synthetic Open Bandit data, with a misspecified model and large drift, it was heavily biased, and the estimator without reuse was the better choice. Second, checks on the anchor or a held-out window could not repair this: at per-period sample sizes the reuse bias is not observable, an information limit that binds any method validating reuse on the current period alone. Third, gradual drift violates the piecewise assumption and favours methods that model a trend. Fourth, our heterogeneity-adjusted interval undercovers when bias remains, and the certified alternative is wide. Finally, on real clicks the natural drift was below what the anchor can observe, so that test shows O-DRIFT is safe when there is little to correct (within 5% of the best baseline), not that it helps; logs with larger natural reward drift remain open.

Two directions follow: reward models flexible enough to make (A4) plausible, so that reuse is safe when attempted, and reporting for each estimate how much unobservable drift would change a deployment decision.

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